

Hotel Management System

PLATFORM – VITYARTHI

DOMAIN – PYTHON PROGRAMMING &
SOFTWARE DEVELOPMENT

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1 . INTRODUCTION :-

Hi every one this project solves the major problem faced by todays hotels, restaurant, resorts and etc.

My main goal to create this project is to create a user friendly interface in python for the above business models as this will help them in maintaining paper less information's which are the most important for them to have with them in order to maintain a proper record of their business with out facing any losses in their business.

The project doesn't requires any kind of heavy software or high specifications in order to run any one can run it with basic requirements.

This program is majorly developed for the day to day use of a small organizations.

2. PROBLEM STATEMENT :-

- This project mainly removes the human error in calculating the bills during the checkout.
- This project removes the risk of double booking for a single room due to which the customer doesn't have to be worried for the miss communication occurred in between the staff member and guest.
- The room service orders are the one which often gets misplaced as they are written on paper and this program helps them to avoid this problems.
- The records created by this programs are going to stay for ever with the user as they are in the form of soft copy where as the traditional way of maintaining records which won't survive for long period of time
- The main object is to remove human error from the tasks performed.

3. FUNCTIONAL REQUIREMENTS :-

1. Customer registration in customers.py :-

- The program collects customer details like unique id, full name, contacts number, aadhaar number, and city.
- Prevent duplicates of registration of same customers id.
- Allow the staff to comfortably see the status of the room.

2. Room inventory in rooms.py :-

- This module is mainly dedicated for displaying the types of rooms available for check-in.
- This also shows real time occupancy status and price of the room per night.

3. Check in operations in booking.py :-

- This validates that the guest is registered before check in.
- Checks whether the room chosen is currently free or not.

- The program also takes input for the no of days of stay
- After every check out the room occupied status is updated to available.

4. In room restaurant service in restaurant.py :-

- First it checks whether the room chosen for orders is it occupied or not then proceeds for the further process.
- Display the menu provided by the restaurant.
- Allows continuous item ordering with quantity.
- It automatically generates food bill and then merge it with the total bill.

5. Bill at the time of check out in billing.py :-

- Computes total room charges.
- Adds all the bill and creates a detailed bill.
- The bill is then printed in the terminal.

6. Customer check out status and history:-

- The checkout details in checkout_record.txt

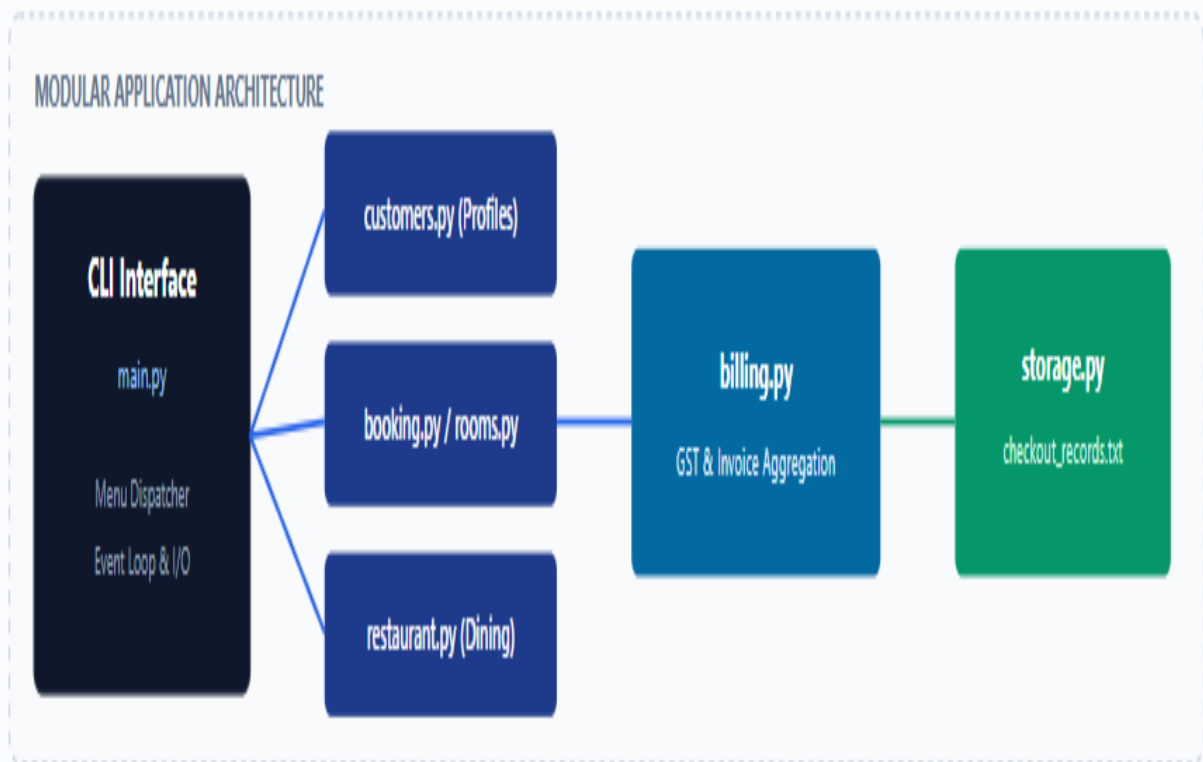
4. NON – FUNCTIONAL REQUIREMENTS:-

- Clear input and out put structure with accurate instructions for ease in operations.
- There is no need for the installation of any external modules.
- The time complexity is really less for the program .
- The program is written in a way that the error performed by the user can be understood by the program and then provide the proper instructions.

5. SYSTEM ARCHITECTURE :-

The project is all linked to each other and all the modules are then controlled with the main.py file .

SYSTEM ARCHITECTURE & MODULE INTERACTION FLOW

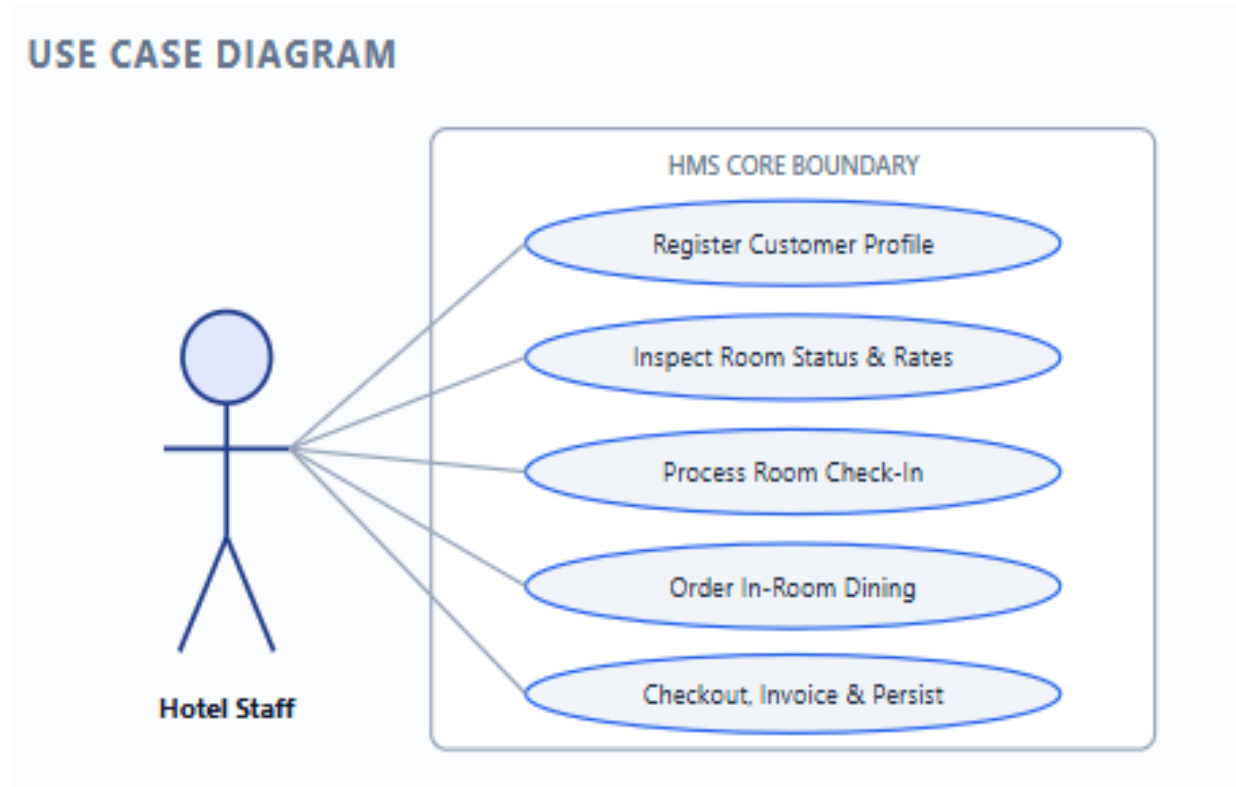


Module responsibilities:

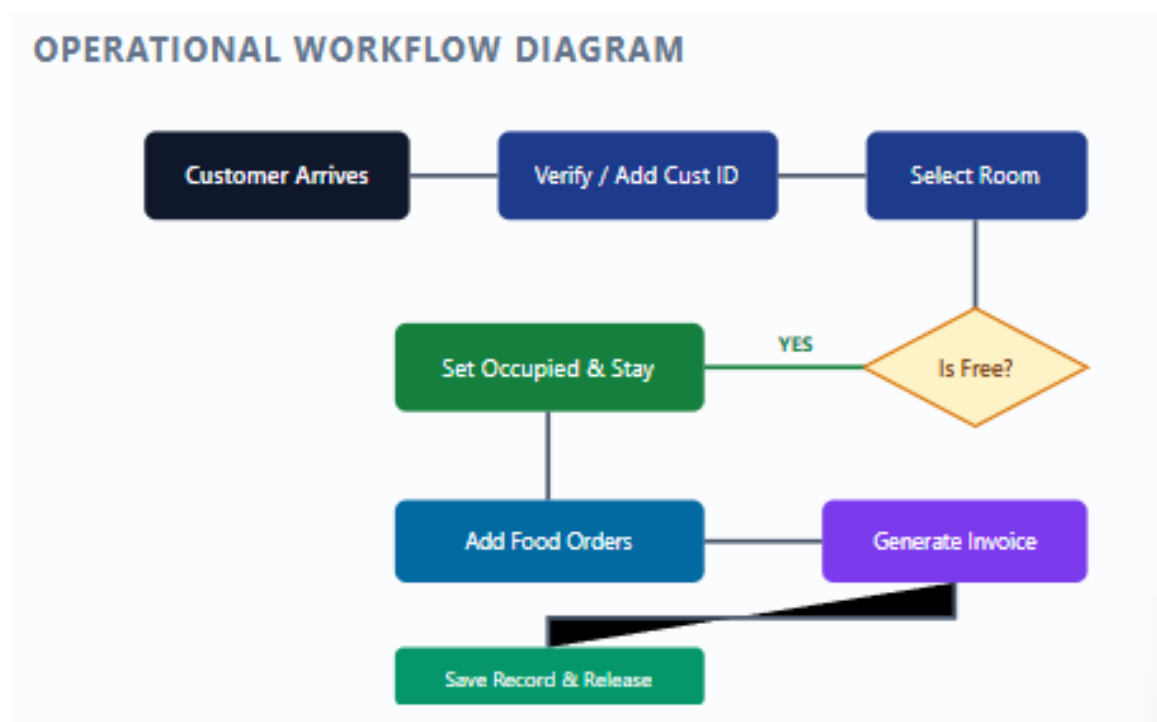
1. `main.py` runs the interactive while loop, captures choices, handles menu input validation, and more.
2. `room.py` holds room type , night rate , live status , and the active stays registry.
3. `customers.py` manages guest records, aadhaar number, and profile lookups.
4. `booking.py` manages check in and check out status.
5. `restaurant.py` contains the food menu and attaches dining charges to active stays.
6. `Billing.py` computes total amount and generates bill.
7. `Storage.py` appends checkout summaries to a file.

6.DESIGN DIAGRAMS :-

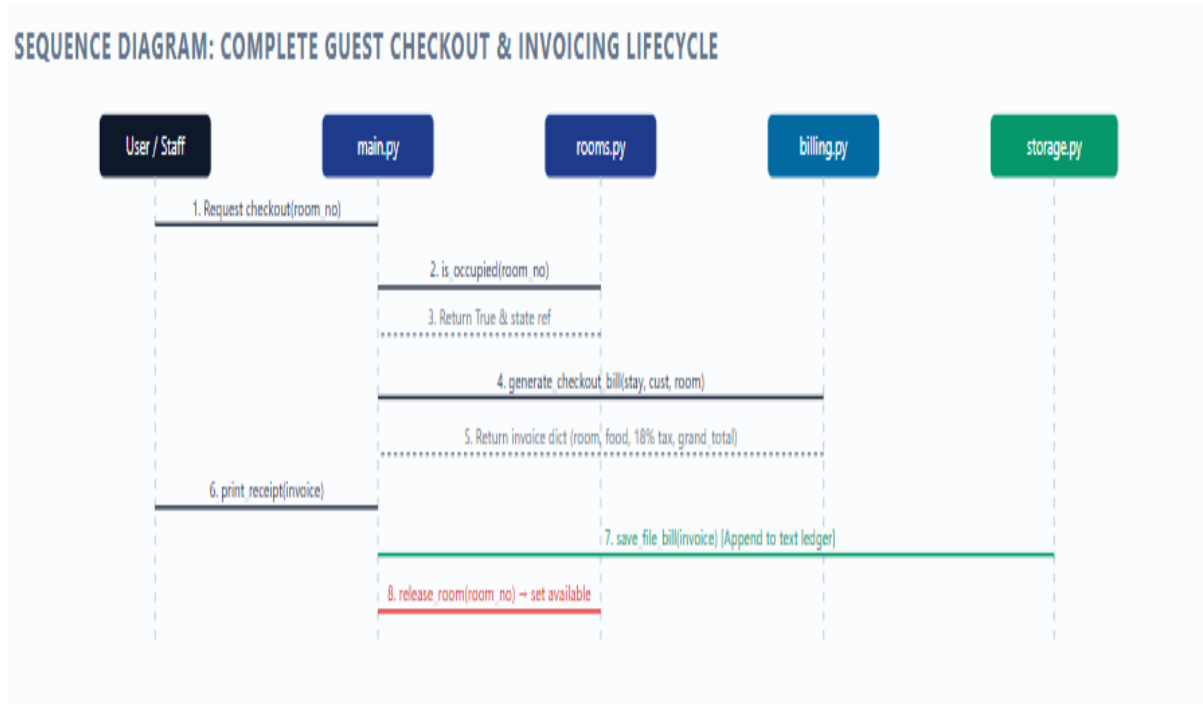
- Use Case Diagrams:



-Operational workflow diagram:

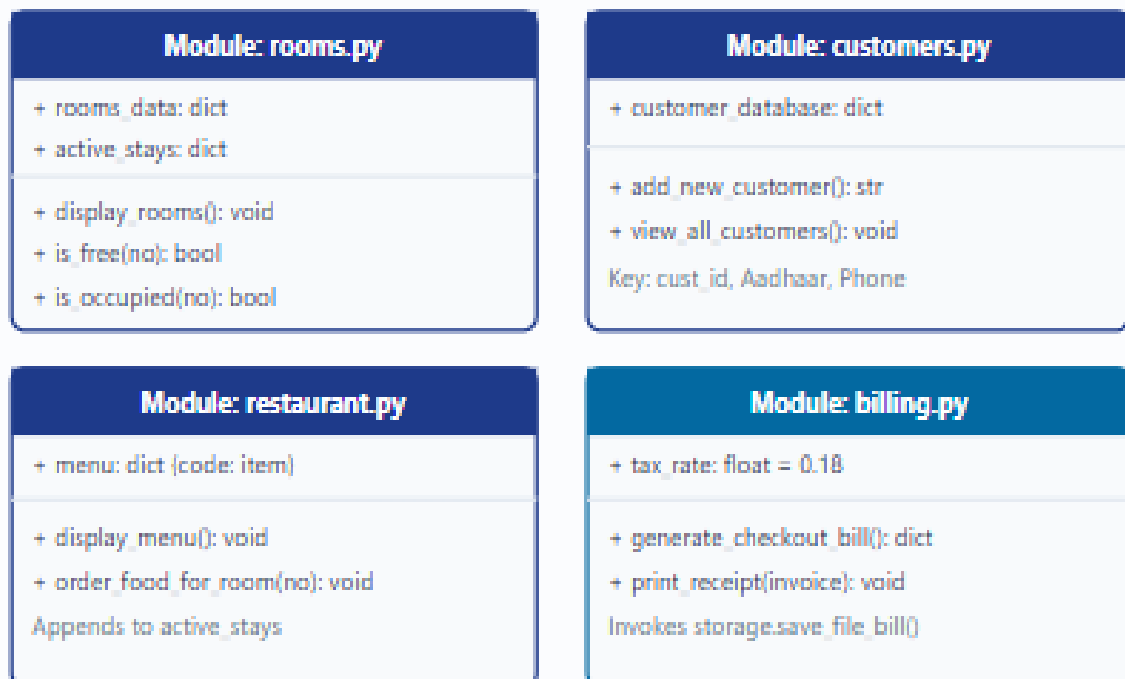


-Sequence diagram:



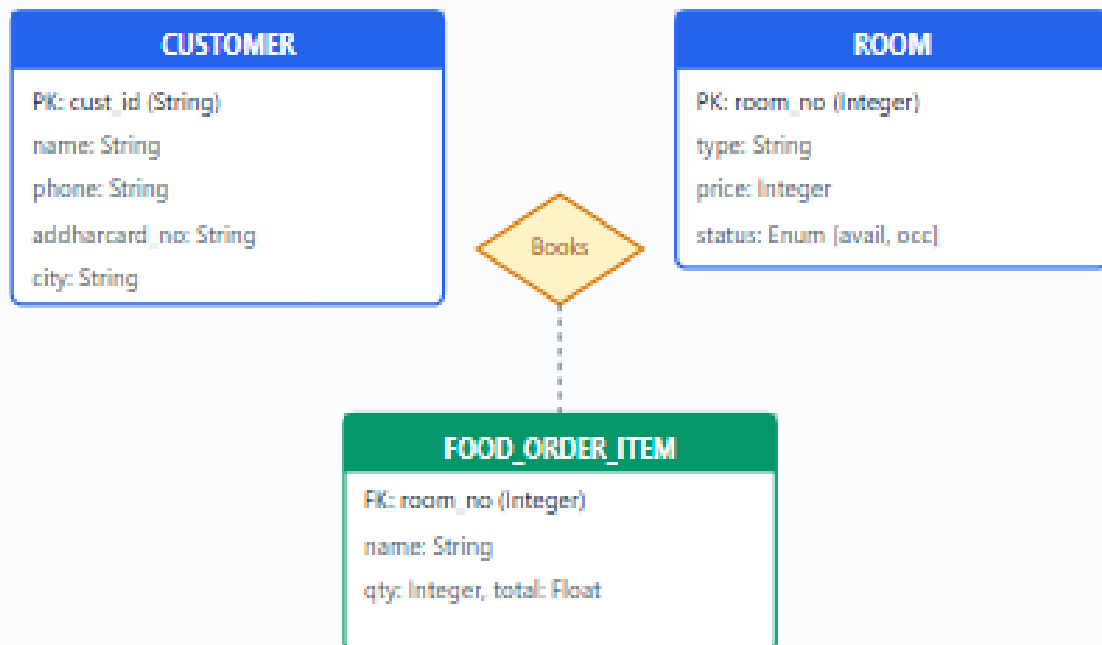
-Class:

CLASS / COMPONENT ENTITY DIAGRAM



-Data Storage:

DATA STORAGE & ENTITY RELATIONSHIP (ER)



7. Design Decisions & Rarionale:-

1. I used dictionary because the room are very limited and for a small project like this adding unnecessary efforts is waste of time.
2. I used append statements because it creates a less time and space complexity in the project and it is more efficient for this project.
3. I created different billing modules because this helps in bringing a clarity in the info related to

different modules and also reduces the confusion during project development.

4. I have also added the tax calculation because in country like india it is really important for the people who run business like this.

8. IMPLEMENTATION DETAILS:-

KEY CODE WALKTHROUGH

- Tax & Invoice Generation:

```
#module 5 = billing

tax_rate = 0.18 # 18 % gst

def generate_checkout_bill(room_no, room_info, stay_info, cust_info):
    days = stay_info["days"]
    price_per_night = room_info["price"]
    room_total = price_per_night * days

    # calculation of total restaurant bill
    food_total = 0
    for order in stay_info["food_orders"]:
        food_total = food_total + order["total"]

    subtotal = room_total + food_total
    tax = subtotal * tax_rate
    grand_total = subtotal + tax

    invoice = {
        "cust_name": cust_info["name"],
        "cust_phone": cust_info["phone"],
        "room_no" : room_no, "room_type": room_info["type"], "days": days,
        "rate": price_per_night, "food_orders": stay_info["food_orders"], "tax": tax, "room_total": room_total, "food_total": food_total,
        "grand_total": grand_total
    }
    return invoice
```

- Preventing Double Booking:

```
def is_free(room_no):  
    if room_no in rooms_data:  
        return rooms_data[room_no]['status']=="available"  
    return False  
  
def is_occupied(room_no):  
    if room_no in rooms_data:  
        return rooms_data[room_no]['status']=="occupied"  
    return False
```

- Durable Record saving:

```
def save_file_bill(invoice):  
    file = open("checkout_records.txt", "a")  
    file.write(  
        f"customer: {invoice['cust_name']} room: {invoice['room_no']} ({invoice['room_type']}) days: {invoice['days']}",  
        f"room cost: rs.{invoice['room_total']} food cost: rs.{invoice['food_total']} grandtotal: rs{invoice['grand_total']:.2f}"  
    )  
    file.close()  
    print("invoice save to checkout records")
```

9. SCREENSHOTS:-

- Main Menu:

Hotel Management System

1. Register a new customer
2. view registered customers
3. room prices
4. check-in
5. order food to room
6. checkout and billing
0. exit

enter process no.(1-6,0(for exit)):

- Guest Registration:

--- Register New Customer ---

Enter unique id for customer: c101

Enter full name: Bhavesh

Enter phone number: 3456278541

enter addhar no: 4321-8765-1092

enter city: Bhaopal

customer 'Bhavesh' registation completed secquerly

- Check in:

enter customer id: c101

Available Rooms & Rates

Room No Room Type Price Per Night Status

1001 tent Non-AC Rs.1000 available

2002 tent AC Rs.2500 available

3003 tree house AC Rs.4000 available

4004 bambu house AC Rs.8500 available

5005 presidential suite Rs.10000 available

enter the room no to book: 4004

no of days of stay: 5

the customer is checked in room no 4004 for mr/ms. Bhavesh.

- Resturant services:

enter room no to order food4004

Restaurant Menu

code item name price (rs)

1 paneer tokka Rs. 220

2 chicken tikka Rs. 280

3 dal makhani Rs. 200

4 butter chicken Rs. 320

5 tanduri roti Rs. 35

6 butter naan Rs. 45

7 cold drinks Rs. 50

enter food code to order (or 0 to finish): 7

enter quantity for 'cold drinks': 2

the bill is added to the rooms total bill _

- Checkout Invoice:

enter room no for check out4004

checkout

customer name : Bhavesh
phone number : 1649119986
room number : 4004
duration : 5
room charges : rs. 42500

resturant orders
cold drinks * 2 = rs.100.00
food total :rs. 100.00

grand total : rs.50268.00

invoice save to checkout records
the room no 4004 is now availabe for booking

- Storage :

customer: bn room: 5005 (presidential suite) days: 1 room cost: rs.10000 food cost: rs.100 grandtotal: rs11918.00customer: Bhavesh room: 4004

10. TESTING APPROACH:-

TEST ID	SCENARIO	INPUT	EXPECTED BEHAVIOR	OBSERVED RESULT	STATUS
1	REGISTER CUSTOMER A DUPLICATE ID	C101	Displayed customer id	Rejected Cleanly	passed
2	Check in unregistered	C999	No customer found	Return to main menu	passed
3	Check in to an occupied room	Room 2002	Room occupied	No double Booking	passed
4	Invalid day to stay	-3 or abc	Only positive integers	no step taken	passed
5	Order food to an empty room	Room unoccupied	Room not checked in	No food order	passed
6	Invalid restaurant item	Code 99	invalid	Code skipped	passed
7	Unoccupied room checkout	Room free	Display room is ready	Aboted ckeckout	passed
8	Record check	Perform check out	Text appended	All records present	passed

11. CHALLENGES FACED:-

- I was commenting a lot of spelling errors.
- I was also doing a lot of syntex error.
- Consistent error , lost in correcting errors.
- Felling frustrated due to small errors like no assigned variable found and etc.

12. LEARNING & KEY TAKE AWAYS:-

- As this is my first python project I learnt various ways of data structures.
- I learnt how to create a module and how to make it functional.
- I learnt how to control my frustration.
- Learnt how to store data in different forms.

13. FUTURE ENHANCEMENT:-

- I will like to create a html inter face for this program.
- I will also include discount based calculation and input for percentage of discount.
- I will like to simplify the project in various different ways.

14. REFERENCES:-

- Mainly my professor Dr. Preetum Suman Sir.
- The online learning platform Vityarthi.
- YouTube .

Thank You